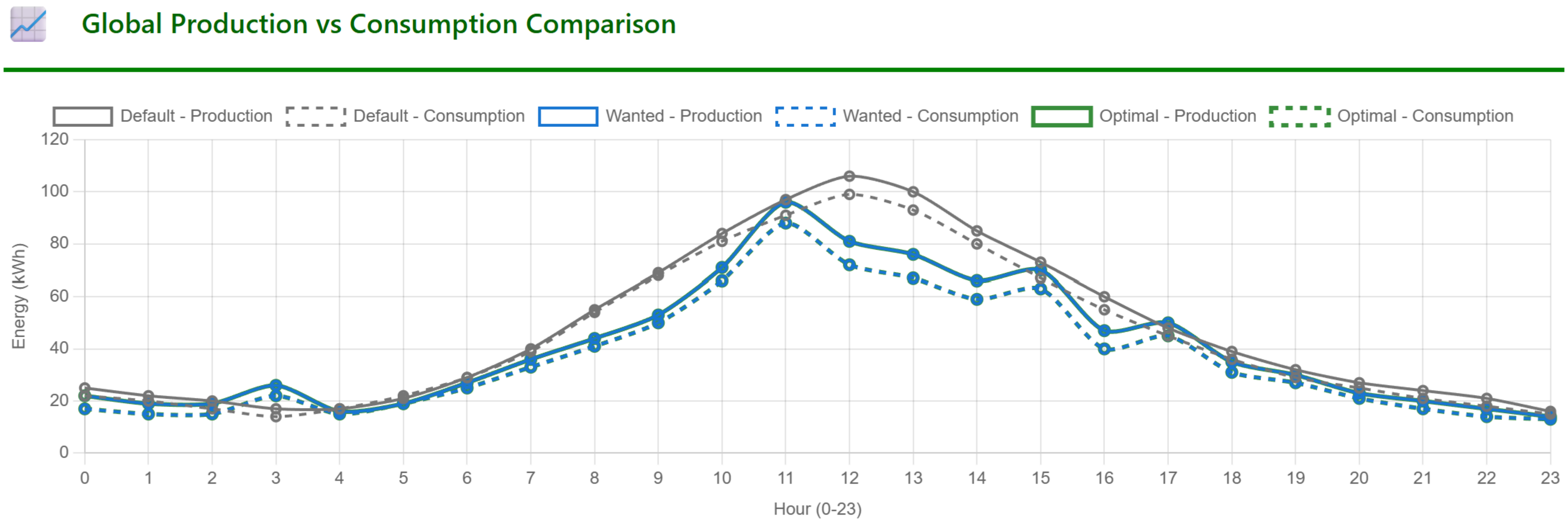
PowerHive

Presentation of 3° Sprint

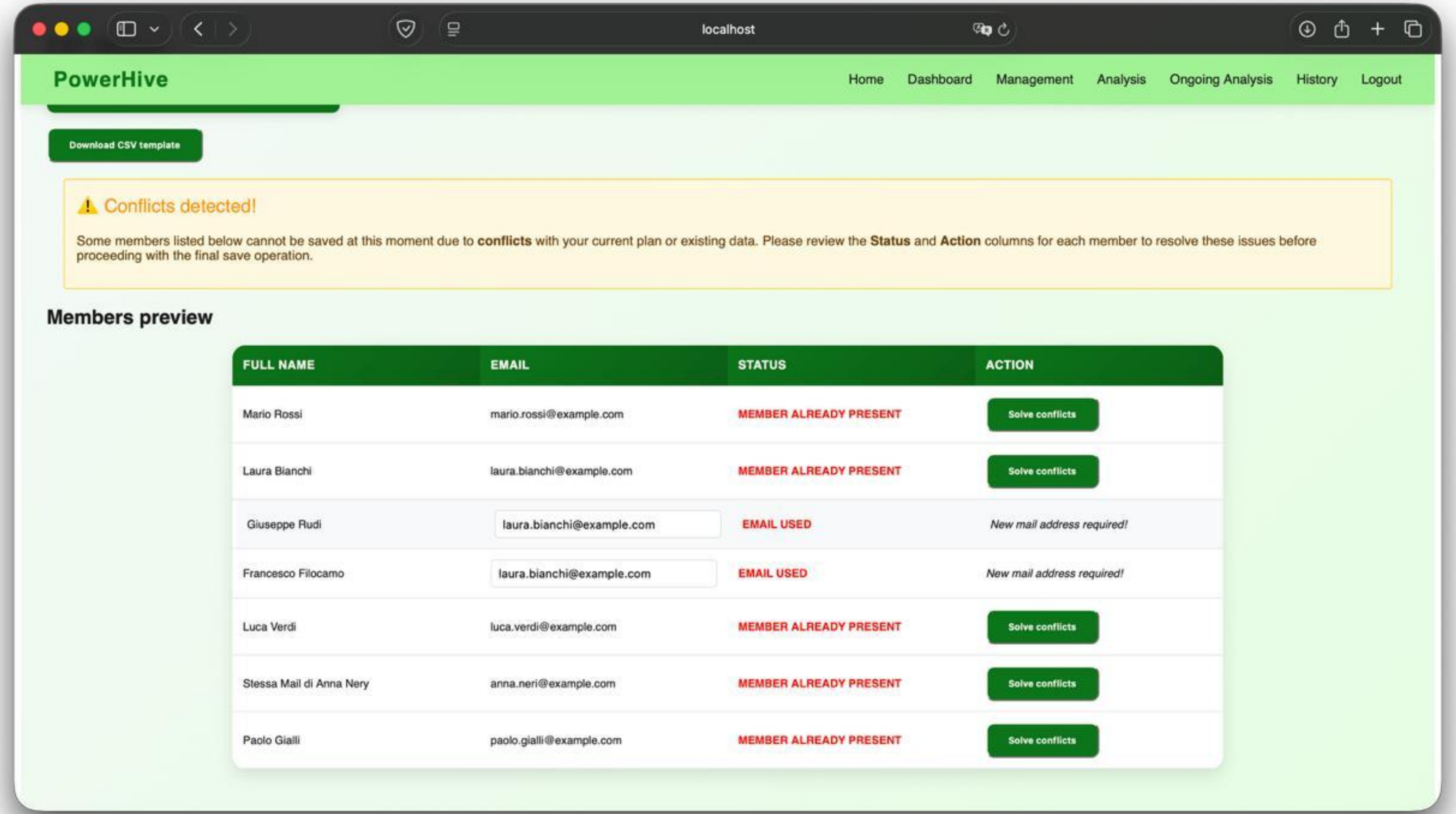
Sprint Goal

Evaluate and optimize battery allocation to maximize shared energy and economic benefits, improve plan management with conflict handling, and enhance analysis history, saving and visualizing past results to enable reuse.

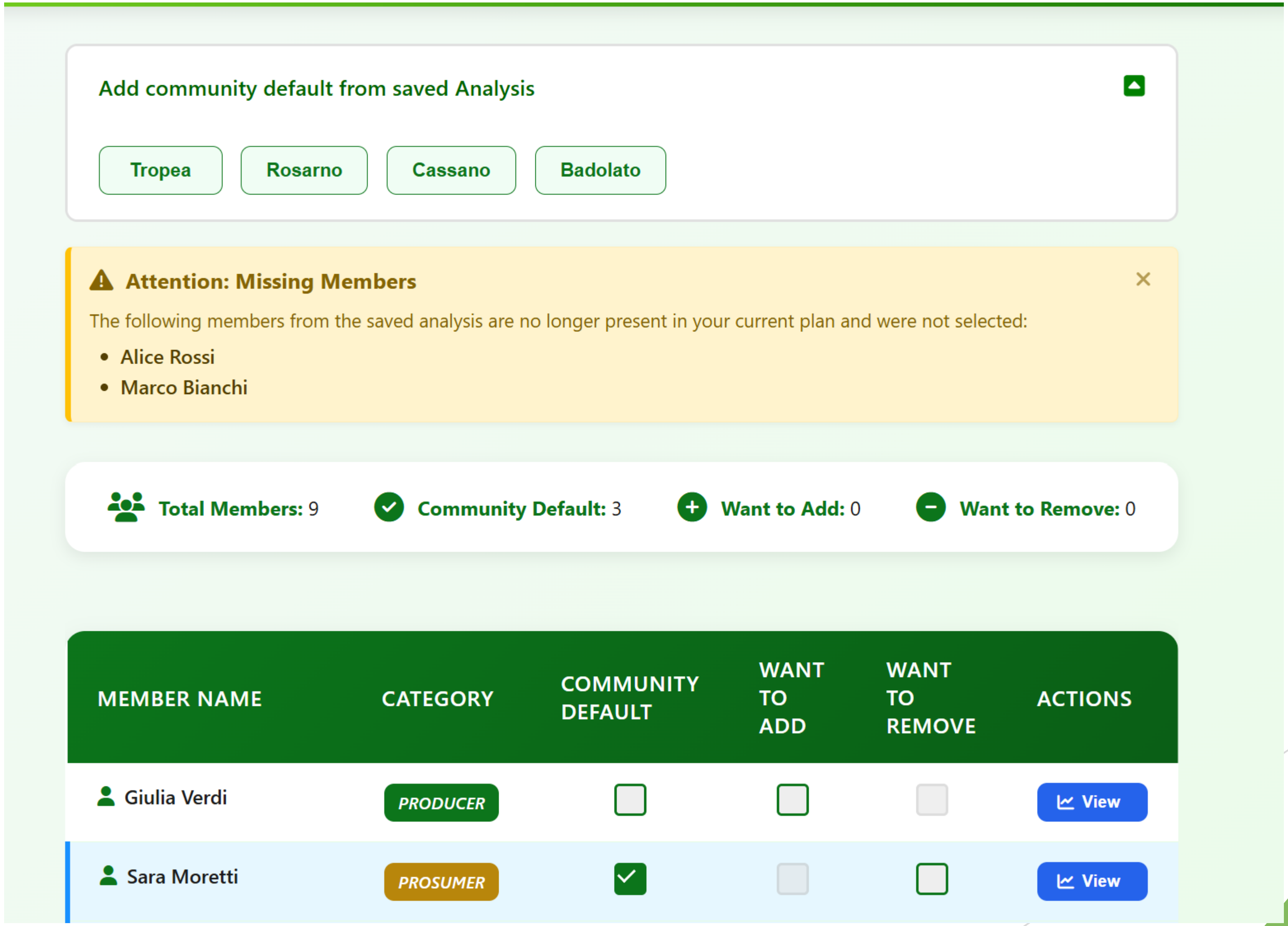
- Refinement of UI/UX and Better Graph Comparations



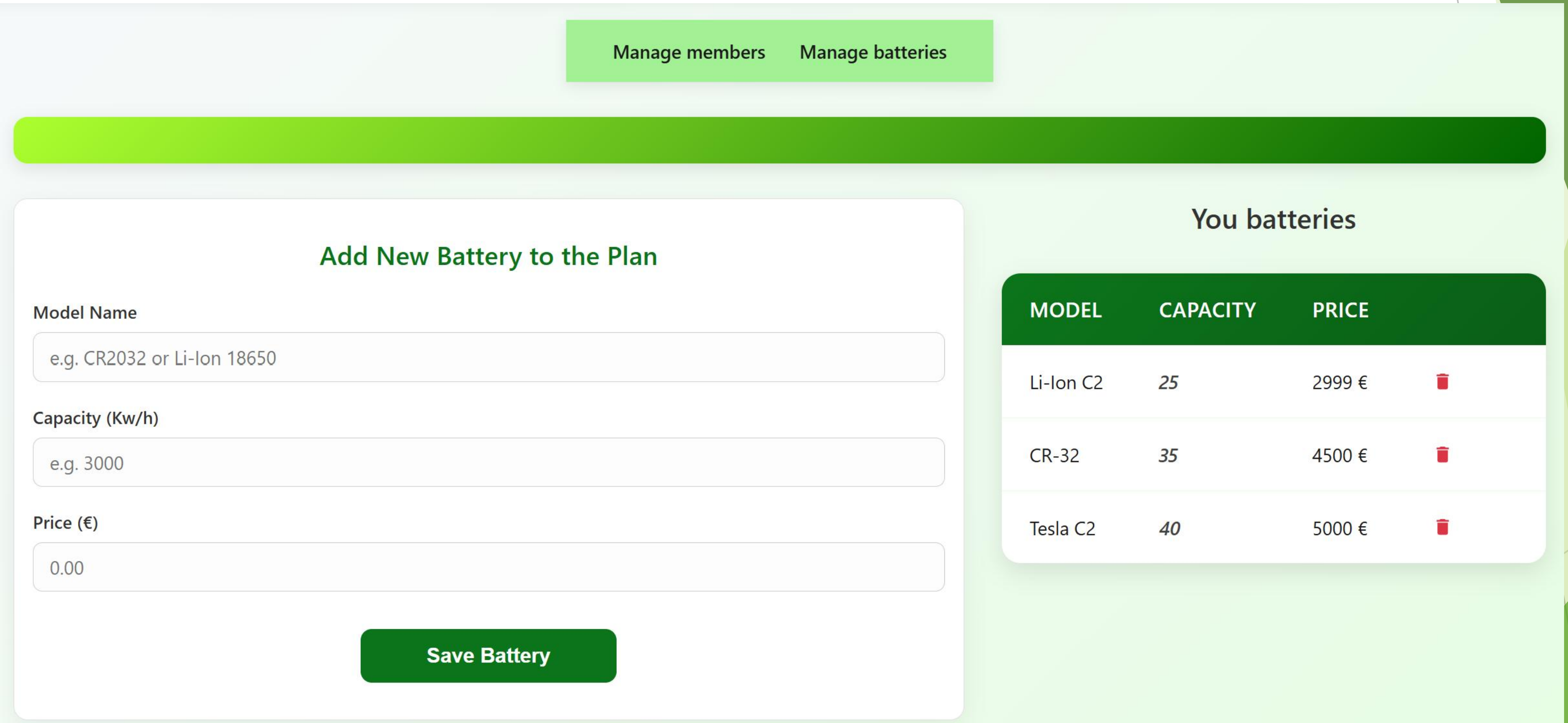
- Plan Conflits Management



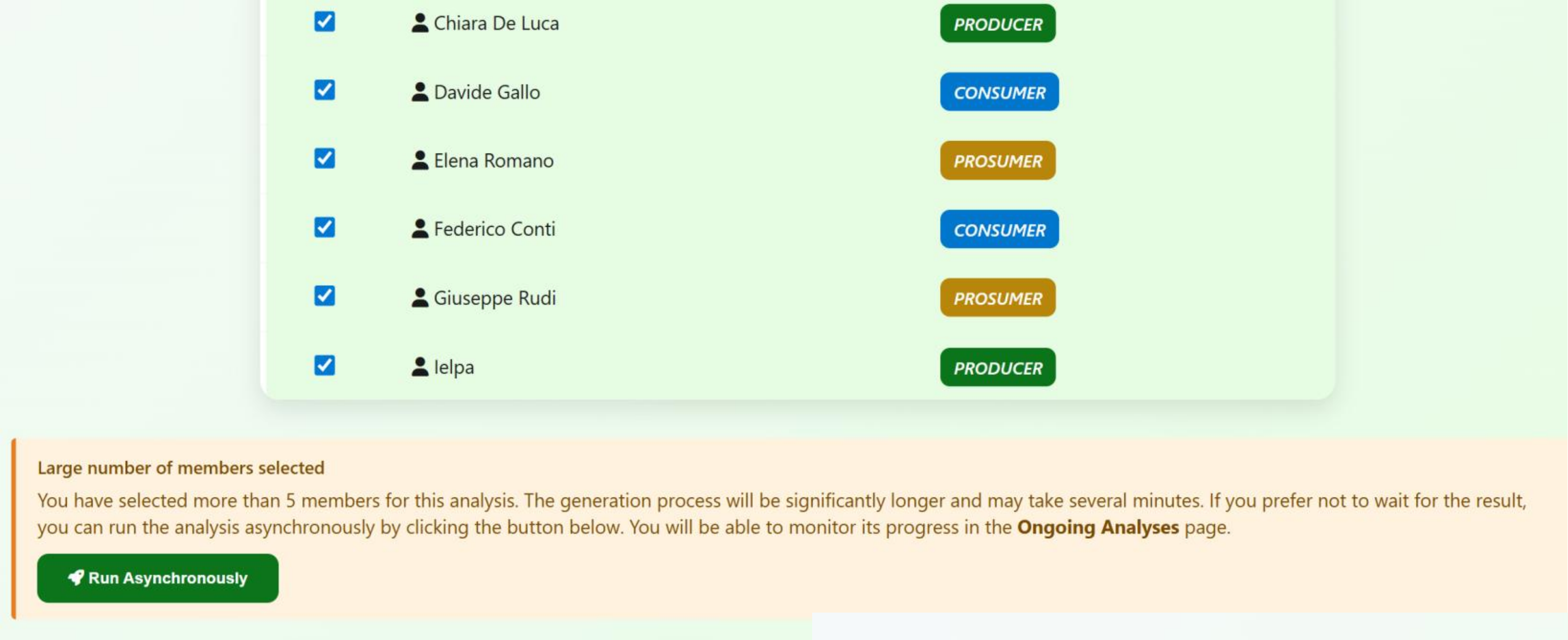
- Reuse of previously created energy communities for the next analyses

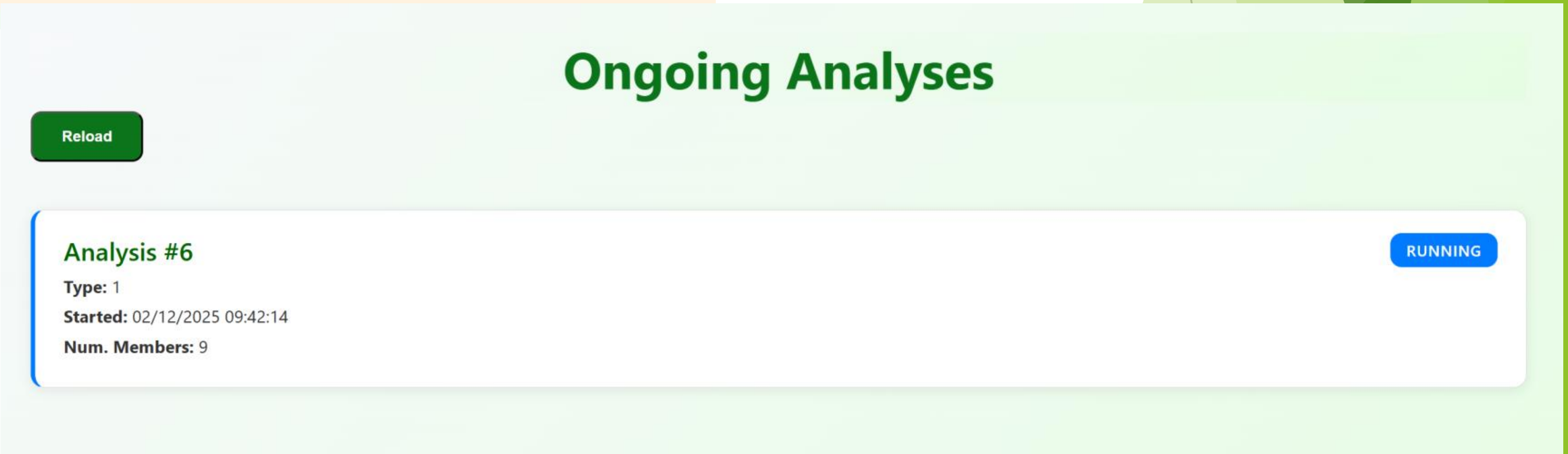


- Batteries Management



- Possibility to run the «Profile Analysis» in Asynchronous Mode





- Fourth Analysis:
Optimal Batteries Assignments

1

Pylontech

Capacity: 10 kWh

Price: **1200 €**

2

Deye RW-M6.1-B

Capacity: 15 kWh

Price: **2000 €**

3

Tesla CF1000

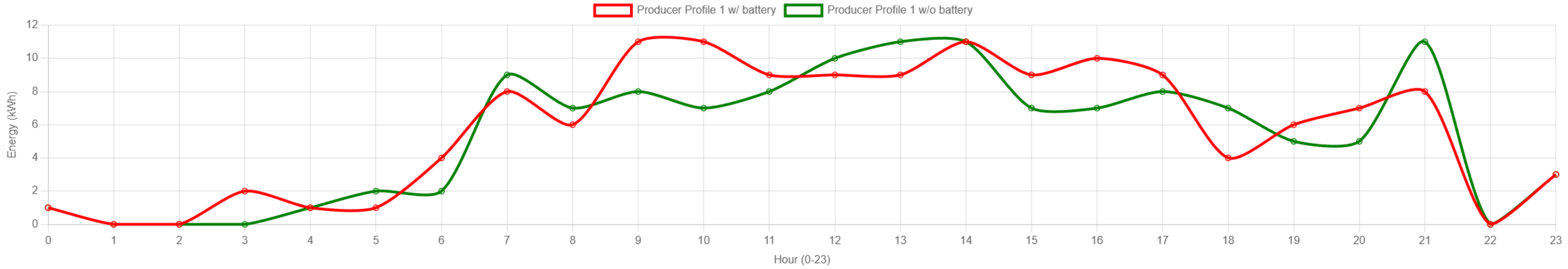
Capacity: 30 kWh

Price: **4500 €**

Optimal Battery Assignment

Mario Rossi 2 PRODUCER

Production Graph



- Fourth Analysis: Overall Financial Analysis

Costs Analysis

Comparison between the costs with/without batteries and visualisation of time to return of investment.

Annual cost without batteries
3,066 € / year

Annual cost with batteries
1,314 € / year

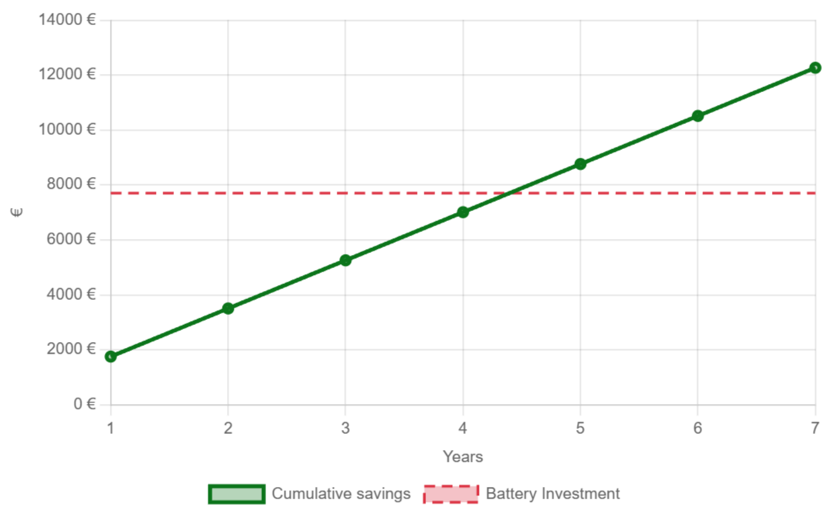
Annual savings thanks to batteries
1,752 € / year

Battery investment
7,700 €

Payback time
4.4 years

Is it convenient to invest in batteries?
YES

Return of Investment



The graph shows the **cumulative savings month by month** compared to the **initial investment** in the battery. The point where the curves intersect is the **estimated payback period**.

- Ability to save the first three types of analyses

Legend: 1 Profile Analysis 2 Optimal Community Formation 3 Community Adjustment Optimization

ANALYSIS NAME	TYPE	CREATION DATE	ACTION
Tropea	2	02/12/2025 09:33:25	Visualize
Rosarno	2	28/11/2025 14:57:19	Visualize
Cassano	2	28/11/2025 14:55:17	Visualize
SanGineto	1	28/11/2025 14:47:27	Visualize
Prova3	3	20/11/2025 17:39:47	Visualize

Next Improvements

- Possibility to add the asynchronus mode in all analyses
- Add the possibility to save the «Optimal Battery Assignment» analysis.
- Reuse the previously saved analyses for the «Optimal Battery Assignment» analysis.
- General improvements
- Possibility to choose the batteries and members to perform «Optimal Battery Assignment» analysis.
- Possibility to use a fixed number of already owned batteries.

Stakeholders requests or ideas ?